```
import pandas as pd
```

```
df=pd.read_csv('/content/gender_submission.csv')
df
```

	PassengerId	Survived
0	892	0
1	893	1
2	894	0
3	895	0
4	896	1
...	...	...
413	1305	0
414	1306	1
415	1307	0
416	1308	0
417	1309	0

418 rows × 2 columns

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0

dtype: int64

```
from sklearn.preprocessing import LabelEncoder
encoder=LabelEncoder()
for col in df.select_dtypes(include='object').columns:
  df[col]=encoder.fit_transform(df[col])
```

```
x=df.iloc[:,0]
y=df.iloc[:,1]
```

```
import numpy as np
```

```
x=np.array(x).reshape(-1, 1)
y=np.array(y).reshape(-1, 1)
```

```
x
```

```
array([[ 892],
       [ 893],
       [ 894],
       [ 895],
       [ 896],
       [ 897],
       [ 898],
       [ 899],
       [ 900],
       [ 901],
       [ 902],
       [ 903],
       [ 904],
       [ 905],
       [ 906],
       [ 907],
       [ 908],
       [ 909],
       [ 910],
       [ 911],
       [ 912],
       [ 913],
       [ 914],
       [ 915],
       [ 916],
       [ 917],
       [ 918],
       [ 919],
       [ 920],
       [ 921],
       [ 922],
       [ 923],
       [ 924],
       [ 925],
       [ 926],
       [ 927],
       [ 928],
       [ 929],
       [ 930],
       [ 931],
       [ 932],
       [ 933],
       [ 934],
       [ 935],
       [ 936],
       [ 937],
       [ 938],
       [ 939],
       [ 940],
       [ 941],
       [ 942],
       [ 943],
       [ 944],
       [ 945],
       [ 946],
       [ 947],
       [ 948],
       [ 949],
```

```
y
```

```
array([[0],
       [1],
       [0],
       [0],
       [1],
       [0],
       [1],
       [0],
       [1],
       [0],
       [0],
       [0],
       [1],
       [0],
       [1],
       [1],
       [0],
       [0],
       [1],
       [1],
       [0],
       [0],
       [1],
       [0],
       [1],
       [0],
       [1],
       [0],
       [0],
       [0],
       [0],
       [0],
       [1],
       [1],
       [0],
       [0],
       [1],
       [1],
       [0],
       [0],
       [0],
       [0],
       [0],
       [1],
       [1],
       [0],
       [0],
       [0],
       [1],
       [1],
       [0],
       [0],
       [1],
       [1],
       [0],
       [0],
       [0],
       [0],
```

```
from sklearn.model_selection import train_test_split
```

```
x_train,x_test,y_train,y_test=train_test_split(
    x,y,test_size=0.2,random_state=42
)
```

```
x_train
```

```
array([[1228],
       [ 923],
       [ 976],
       [1179],
       [1209],
       [1103],
       [ 986],
       [1281],
       [1219],
       [ 897],
       [ 937],
       [1032],
       [1037],
       [1295],
       [1250],
       [ 908],
       [1278],
       [1016],
       [1217],
       [ 895],
       [ 910],
       [1024],
       [ 952],
       [1090],
       [1224],
       [ 955],
       [1157],
       [1085],
       [1084],
       [1158],
       [1088],
       [1008],
       [ 918],
       [ 899],
       [1142],
       [1138],
       [1000],
       [ 993],
       [1202],
       [ 921],
       [1087],
       [1282],
       [1010],
       [1163],
       [1006],
       [1203],
       [1023],
       [1049],
       [1044],
       [1263],
       [1291],
       [1164],
       [1292],
       [1272],
       [1001],
       [1068],
```

```
       [1258],
```

```
y
```

```
array([[0],
       [1],
       [0],
       [0],
       [1],
       [0],
       [1],
       [0],
       [1],
       [0],
       [0],
       [0],
       [1],
       [0],
       [1],
       [1],
       [0],
       [0],
       [1],
       [1],
       [0],
       [0],
       [1],
       [0],
       [1],
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       [1],
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       [0],
       [1],
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       [1],
       [1],
       [0],
       [0],
       [1],
       [1],
       [0],
       [0],
       [0],
       [0],
```

```
y_train
```

```
array([[0],
       [0],
       [0],
       [0],
       [0],
       [0],
       [0],
       [0],
       [0],
       [0],
       [0],
       [1],
       [0],
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       [0],
       [0],
       [0],
       [0],
       [1],
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       [0],
       [1],
       [0],
       [0],
       [1],
       [0],
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       [0],
       [1],
       [1],
       [0],
       [0],
       [1],
       [0],
       [0],
```

```
from sklearn.linear_model import LogisticRegression
```

```
model=LogisticRegression(class_weight='balanced')
```

```
model.fit(x_train,y_train)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:1408: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the s
  y = column_or_1d(y, warn=True)
▾ LogisticRegression
LogisticRegression(class_weight='balanced')
```

```
y_pred=model.predict(x_test)
```

```
from sklearn.metrics import accuracy_score,confusion_matrix,precision_score,recall_score,f1_score
```

```
accuracy=accuracy_score(y_test,y_pred)
print("Accuracy:",accuracy)
```

```
Accuracy: 0.5238095238095238
```

```
from sklearn.metrics import confusion_matrix
```

```
cm=confusion_matrix(y_test,y_pred,labels=[1,0])
print("Confusion Matrix:")
print("TP FP")
print("FN TN")
print(cm)
```

```
Confusion Matrix:
TP FP
FN TN
[[18 16]
 [24 26]]
```

```
precision = precision_score(y_test, y_pred, zero_division=0)
print("Precision:",precision)
```

```
Precision: 0.42857142857142855
```

```
recall = recall_score(y_test, y_pred, zero_division=0)
print("Recall:",recall)
```

```
Recall: 0.5294117647058824
```

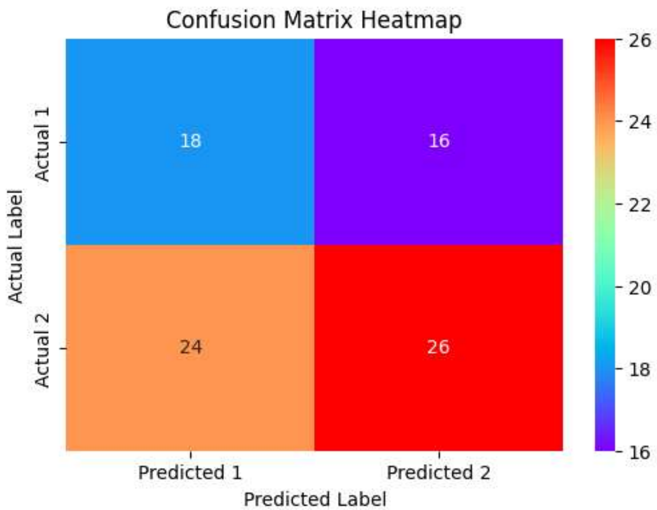
```
f1=f1_score(y_test, y_pred, zero_division=0)
print("F1 Score:",f1)
```

```
F1 Score: 0.47368421052631576
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
cm=confusion_matrix(y_test,y_pred,labels=[1,0])
```

```
plt.figure(figsize=(6,4))
sns.heatmap(cm,annot=True,fmt='d',cmap='rainbow',
            xticklabels=['Predicted 1','Predicted 2'],
            yticklabels=['Actual 1','Actual 2'])
plt.xlabel('Predicted Label')
plt.ylabel('Actual Label')
plt.title('Confusion Matrix Heatmap')
plt.show()
```




```
from sklearn.metrics import roc_curve,auc
```

```
y_prob=model.predict_proba(x_test)[:,1]
```

```
fpr,tpr,thresholds=roc_curve(y_test,y_prob)
```

```
roc_auc=auc(fpr,tpr)
print("AUC:",roc_auc)
```

```
AUC: 0.5229411764705882
```